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EXAMINING THE IMPACT OF SKILL MISMATCH ON LABOUR FORCE PARTICIPATION: AN EMPIRICAL ANALYSIS

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Abstract: This paper meticulously investigates the influence of skill and educational mismatches on labour force participation. By employing sophisticated econometric models on panel data, the analysis uncovers nuanced relationships between mismatches, hours worked, and earnings. The study offers valuable insights into the significant discrepancies in labour force participation arising from over-education, under-education, over-skilling, and under-skilling, thus providing critical information for future policy design and labour market strategies to foster more balanced and equitable outcomes.

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STATEMENT OF ORIGINALITY

This document is written by Odud Bin Yousuf who declares to take full responsibility for the contents of this document. I declare that the text and the work presented in this document are original and that no sources other than those mentioned in the text and its references have been used in creating it. UvA Economics and Business is responsible solely for the supervision of completion of the work and submission, not for the contents.

1. INTRODUCTION

"Economic slowdown likely to force workers to accept lower quality jobs," declared the news headline of the International Labour Organisation's World Employment and Social Outlook: Trends 2023 report (WESO Trends). This report by Kühn and Ernst (2023) suggested that many workers might have to accept lower quality jobs, often at very low pay and sometimes with insufficient hours. The WESO Trends suggested that the economic slowdown was primarily caused by emerging geopolitical tensions, the Ukraine conflict, uneven pandemic recovery, and ongoing issues in global supply chains. This slowdown may cause mismatches in the labour market, therefore impacting labour force participation. Richard Samans, Director of the ILO's Research Department and coordinator of the report by Kühn and Ernst (2023), stated that this impact on labour force participation is of significant concern.

The significance of labour force participation, both as an explanatory factor for unemployment and as an independent economic variable, has been underpinned by several papers. For instance, Brewer (2001) used hours-contingent on payments as a proxy for labour force participation when assessing the tax reform policies for families in the U.K. The econometric analysis done in papers like these and Hoynes (1996), Keane and Moffitt (1998), Blundell et al. (2000), Van Soest et al. (2002) elaborate on the importance of labour force participation, and the variables affecting/quantifying it, when considering labour markets and their supply. A few important factors that affect the labour force participation of workers are their education level, their level of satisfaction with the job they have and how well their skill level matches the skills required in their current job, as mentioned by Allen and Van der Velden (2001).

This thesis will therefore focus on skill matches and subsequent educational matches, given that skills are predominantly acquired through education, as these factors have been identified as influential determinants of labour force participation in studies such as Hersch (1991), Topel (1986), Viscus (1979) and Tsang and Levin (1985). Moreover, a worker's earnings and weekly working hours directly impact their engagement in the labour force. Several of the aforementioned academic papers employ earnings and working hours as proxies for labour force participation. These papers define a skill mismatch between the worker and their job to occur when the skills that the workers have are not aligned with the skills required from their job. This misalignment could be in terms of the worker being over-skilled (having more skills than the work requires), under-skilled (having less skills than the work requires) or just differently-skilled (where the skills the worker possesses are of a different kind than the skills required of the job). Educational mismatch is defined similarly. Upon investigation of educational mismatch, Topel (1986) and Hersch (1991) found that over-education affects labour turnover while Tsang and Levin (1985) found that over-education affects job satisfaction.

This impact of skill mismatch on labour force participation entails further examination making it the focus of my thesis. The subsequent sections aim to provide a comprehensive exploration of this topic. The second section presents a theoretical framework that builds upon previous research and findings. The theoretical framework utilizes the previous research to define the several ideas and concepts mentioned in this introduction. An extensive breakdown of mismatches in skill and education, and their link to labour force participation is detailed.

The third section uses the theory mentioned to put forth the relevant methodology that can be used and has been used when analysing these trends. This methodology discusses the appropriate estimation methods and models that are used when dealing with panel data. In Section 4, the data and the derived estimates are presented and described, along with a brief discussion of the methodology's role in obtaining these results. These findings serve as the basis for drawing an overall conclusion regarding the research question posed in this thesis. Section 5 encompasses this conclusion, summarizing the entire paper and suggesting potential avenues for future studies to explore.

2. THEORETICAL FRAMEWORK

2.1. *Defining Skills*

Despite the central role played by skill in the labour market, there is limited consensus in the literature regarding its precise definition. According to OECD (2017), skills are defined as encompassing cognitive, non-cognitive abilities, and technical skills. In their work, cognitive skills were described as the ability to comprehend intricate concepts, adapt to the surroundings, acquire knowledge from experience, engage in diverse forms of reasoning, and surmount obstacles through deliberate thought. These cognitive skills include literacy, numeracy, and the capacity to solve abstract problems. On the other hand, non-cognitive skills were characterised as social, emotional, and behavioural aspects, which do not fall under cognitive skills. These non-cognitive skills include work habits, behavioural traits, and physical attributes. Margolis (2014) expanded upon this definition by defining technical skills as a fusion of cognitive and non-cognitive skills employed to accomplish specific tasks.These technical skills are of the job-specific and sector-specific kind.

2.2. *Skill Mismatch and Educational Mismatch*

Schneck (2020) acknowledged these skill definitions and highlighted that these skills are typically acquired through formal education and structured job training programs. They emphasized the need for policymakers who aim to alter the skill composition of the workforce to take into account various measures. These measures included enhancing career guidance services, providing better information on job requirements, and fostering educational policies that align with the demands of the labour market. Their recognition underscored the importance of

strategic interventions to address skill mismatches on the macroeconomic level and promote better skill-to-job matching for improved labour market outcomes.

For the instances in the labour market where this skill mismatch does occur, the 'assignment' theory of Sattinger (1993) makes the critical assumption that skill mismatch is implied by educational mismatch. This assumption, however, has been severely challenged by Green and McIntosh (2007), Mavromaras et al. (2013) and Sánchez-Sánchez and McGuinness (2015), which showed the need to include skill mismatch as a variable of its own as opposed to assume that it's implied when educational mismatch is used. Sattinger (1993) argued that over-educated workers earned less than their well-matched counterparts, and under-educated workers earned more than their well-matched counterparts. Sicherman (1991) and Dolton and Vignoles (2000) stated the same, where Sicherman (1991) found that over-educated workers earned 7% less than their well-matched counterparts, while under-educated workers earned 5% more than the educationally matched workers. Sattinger (1993) concluded that the basic idea behind such results was this: although higher education is assumed to increase productivity, the actual level of productivity realized is determined by the level of mismatch or lack thereof. Consequently, over-educated or/ and over-skilled workers are limited within their jobs, leading to lower productivity.

2.3. *Macroeconomic Mismatch versus Microeconomic Mismatch*

A breakdown of skill mismatch into macroeconomic and microeconomic levels was undertaken by Vandeplas et al. (2019). At the macroeconomic level, they defined skill mismatch as the disparity between the aggregate supply and demand for skills, often examined in the context of a geographical unit such as a country. On the other hand, at the microeconomic level, skill mismatch was characterized by a discrepancy between workers' skill levels and the skill requirements of their respective jobs. This breakdown was utilised by several other papers that are discussed in this section.

Firstly, according to Quintini (2011), skill shortages were observed when employers faced difficulties in recruiting staff with the necessary skills in the labour market at the prevailing wage rates. Secondly, skill surpluses occurred when there was an oversupply of certain skills compared to the level of demand. Thirdly, McGuinness et al. (2017) noted that the term "skill gaps" was frequently employed to describe situations where the skill levels of the existing workforce fell short of meeting the demands of firms. These 3 observations were incorporated into the concept of skill mismatch by McGuinness et al. (2017). The assessment of macroeconomic skill mismatch involved comparing the qualification required in job vacancies to the qualifications of the working age population, serving as a proxy for labour supply, as undertaken by McGuinness et al. (2017) and McGuinness and Redmond (2018).

Macroeconomic indicators of skill mismatch focused on determining whether individuals with specific qualifications or skills were (un)employed, without assessing whether their job aligned to their qualifications as discussed in McGuinness et al. (2017) and McGuinness and Redmond (2018). In contrast, the microeconomic measure of skill mismatch involved comparing the skills or qualifications possessed by an employed worker with the skills or qualifications demanded by their job. This comparison led to workers being categorized as under-skilled or over-skilled in relation to a particular position.

2.4. *Self-reported Mismatch and Realized Mismatch*

McGuinness et al. (2017) further categorized the skill mismatches into two types: self-reported mismatch and realized-matches mismatch. The former pertained to the demand side, where employees were questioned about their possession of the necessary skills to undertake a more demanding job than their current one or whether additional training was required to perform their job satisfactorily. On the other hand, the latter related to the supply side, where the attained values of cognitive skills were compared to the average or median values within the occupation, serving as proxies for job requirements.

Pellizzari and Fichen (2017) conducted a study that emphasized the importance of measuring skill mismatch accurately. By utilizing data from the Survey of Adult Skills (PIAAC), they compared the self-reported approach to the realized-matches approach. The findings revealed a notable difference in the classification of workers' skill mismatches between the two methods. With the self-reported approach, the majority of workers were classified as over-skilled, whereas the realized-matches approach identified a substantial portion of workers as well-matched. This stressed the significant influence of the chosen measurement approach on the characterization of skill mismatches among workers.

One study by Algan and Bartscher (2021) that examined the effects of skill mismatch, utilizing the previously established framework, observed that a discrepancy in skills had adverse effects on labour force participation. Regardless of the specific indicator or approach (i.e. macro- or micro- economic mismatch, or in terms of self-reported vs realized mismatch) used to assess the skill requirements of a job, the study revealed that skill mismatch resulted in a suboptimal allocation of talent and led to productivity losses at the aggregate level.

2.5. *Mismatch and Structural Changes*

When analysing the effect of skill mismatch on labour force participation, the study by Duval et al. (2022) found that most labour markets in advanced economies were tighter than prior to COVID-19. Here, labour market tightness was defined as the phenomenon that exists when there are more jobs available than workers to fill them. They explained this phenomenon via 3

contributing factors: shifting workers' preferences away from certain types of jobs, a mismatch between sectors and occupations, impediments to work re-entry. The authors noted a trend among workers away from low-paying jobs and found that despite a simultaneous increase in job vacancies for contact-intensive jobs, a growing proportion of employees of this sector had either moved to other occupations or left the workforce entirely. Trends such as these indicated a structural mismatch in the labour market.

Restrepo (2015) stated that the global financial crisis of 2007 led to a structural mismatch in the labour market. Structural mismatch occurred when workers' preferences changed, resulting in a mismatch between sectors and occupations. Regarding this structural mismatch, Duval et al. (2022) and Kiss et al. (2022) found no evidence to suggest that it played a significant and durable role in rising job vacancies. On the other hand, Restrepo was a proponent of the idea that such structural changes cause skill mismatches over the mid to long run. Accordingly, Acemoglu et al. (2023) noticed the influx of revolutionary technologies such as Artificial Intelligence and cautioned serious concern about automation displacing workers. The study argued that while automation may lead to short-run job displacement, investing in human capital can create new jobs in the long run.

In conclusion, the theoretical framework presented in this section has shed light on various aspects of skill mismatch in the labour market. The definition of skills encompassed both cognitive and non-cognitive abilities, as well as job-specific or sector-specific proficiency. This section also elaborated upon the role of educational and skill mismatch as the primary factors of interest when analysing labour force participation in an economy. The division of skill mismatch into macroeconomic and microeconomic levels emphasized the difference between the overall supply and demand for skills in the market, and the gap between a worker's abilities and what their job actually requires.

Studies have emphasized the need for strategic interventions to address skill mismatches and promote better skill-job matching for improved labour market outcomes. Additionally, the impact of the chosen measurement approach on the characterization of skill mismatches among workers has been discussed, emphasizing the valuable role of employer input in enhancing the accuracy of information gathered. Due to data limitations regarding the measurement of aggregate demand and supply for skills, this thesis predominantly focuses on analyzing microeconomic self-reported skill mismatch. The available data lacks information on the specific skills required or possessed by sectors or jobs, which presents difficulties in constructing proxies for job requirements and consequently analyzing realized skill mismatch.

3. METHODOLOGY

This paper utilizes data from the Longitudinal Internet Studies for Social Sciences (LISS) which has data available from 2008 to 2022. The Core Study used from LISS is the 'Work and Schooling' study. The 'Background Variables' single wave study of April, 2022 is used as well.

The literature regarding the influence of educational and skill mismatch on earnings, and how mismatch contributes towards explaining the determinants of earnings, is more abundant than literature on mismatch affecting the working hours of workers. More often, cross-sectional data is used to study the aforementioned relations of interest in this paper. This type of data is very susceptible to unobserved heterogeneity characteristics.

To accurately predict the effect of skill mismatch on labour force participation, controlling for all relevant variables that might influence both skill mismatch and labour force participation is key. Therefore, exclusion of factors such as motivation and innate ability while using standard ordinary least squares (OLS) regression models can introduce serious bias into the results. For instance, these omitted control variables could be both correlated with skill mismatch and labour force participation. Individuals with high innate abilities or strong motivation might experience less skill mismatch because they can learn new skills more quickly, and they might be more likely to participate in the labour market because they are driven to succeed. If these variables are excluded from the model, their effects could be incorrectly attributed to the skill mismatch. This would make the skill mismatch appear to have a larger effect on labour force participation than it truly does and result in omitted variable bias. McGuinness and Sloane (2011) attempted to minimize these biases by using latent variable models and probit regression models as well as performing sensitivity analysis and robustness checks on each of the estimates found.

The study conducted by Mavromaras et al. (2013) employed a self-reported microeconomic measurement approach to define over-education and over-skill. In this approach, over-education was defined when workers reported having a higher level of education than required by their job, while over-skill was defined when workers reported possessing skills beyond the requirements of their job. They analysed the effects of over-education and over-skill on wages using panel data from Australia and employed various regression techniques including OLS, Fixed Effects, and Random Effects with Mundlak corrections. The latter two models are commonly employed in panel data analysis to account for unobserved heterogeneity. Mavromaras et al. (2013) concluded that over-education and over-skill exhibit significant adverse effects on wages when using OLS regression. However, the Fixed Effect model and the Random Effect model with Mundlak corrections revealed no significant impact on wages.

Ordinary Least Squares (OLS) regression assumes that all observations are independent, which doesn't hold in the panel data context where multiple observations for the same individual are likely to be correlated. OLS also doesn't account for unobserved individual-specific characteristics that might be correlated with the observed variables and therefore bias the results. The major advantage of using Fixed Effects (FE) and Random Effects (RE) models is that they address the issue of this unobserved heterogeneity.

The FE model is used when we believe that some unobserved individual characteristics (like personal choice of work, innate personal ability, and more) are time-invariant and correlated with the observed variables. These characteristics are captured by individual-specific intercepts (fixed effects). The FE model effectively removes the influence of these unobserved variables by looking at the within-subject variation. This way, the estimates of the observed variables (like skill mismatch) are free of the influence of the unobserved characteristics. However, the FE model cannot estimate the effects of time-invariant observed variables, since it uses within-subject variation.

The RE model, on the other hand, assumes that the unobserved individual-specific effects are not correlated with the observed variables, and treats them as a random sample from a larger population. The advantage of the RE model is that it uses both within-subject and between-subject variation, and so it can also estimate the effects of time-invariant observed variables. It is more efficient than the FE model when the unobserved effects are truly random. Nevertheless, if the assumption of no correlation between the random effects and the observed explanatory variables is not met, it would lead to biased estimates in the Random Effect model.

3.1. *Theoretical models*

McGuinness and Sloane (2011) and Allen and Van der Velden (2001) employed models that bear striking resemblance to the model under investigation in this paper. Hence, these two approaches hold utmost significance for comparison and reference.

The methodological approach adopted by McGuinness and Sloane (2011) utilized cross-sectional graduate cohort data and suggested that the use of their data minimizes the risk of unobserved heterogeneity bias to some degree since respondents have uniform levels of education and labour market experience. While this sample selection method does provide a level of control, it doesn't eliminate all potential sources of bias. There are many other unobserved factors that can influence labour market outcomes, such as innate ability, socio-economic background or personality traits, which aren't controlled for in this approach.

McGuinness and Sloane (2011) regressed a vector of personal and structural characteristics, the over- and under- education dummies, the over- and under- skilled dummies on log wages.

The personal and structural characteristics used were gender, labour market experience, type of degree, level of degree, hours worked, size of the firm, whether the firm is an R&D firm, number of employers of the firm, and whether the firm is a public firm. While hours worked is used as an independent variable in McGuinness and Sloane (2011), this thesis uses it as both a dependent and independent variable.

The measures and variables that are defined and used in this thesis follow very closely the methodology outlined in the papers of Allen and Van der Velden (2001) and McGuinness and Sloane (2011). The few differences in methodologies used will be caused predominantly by the unique data used and the specific relations of interest. This will be made abundantly clear in section 3.2 and 3.3.

3.2. *Variable Definitions*

The LISS panel has yearly longitudinal waves from 2008 to 2022. The definitions of the relevant variables of interest as defined in the code book of LISS are described in Appendix A. Based on the available information in the LISS data, the variables of interest have been re-formulated into a functional form that is both comprehensible and consistent with the previous literature referenced. These are the variables that will be used in any and all mathematical and econometric analysis conducted.

With respect to the data used in this thesis, respondents are defined as over-educated if they indicated that their education was above the level required by their. They are defined as under-educated if they indicated that their education was below the level required by their work. Respondents were educationally-matched if they stated that their education was approximately at the level required by their work. Furthermore, the respondents that answered that their education was of a different kind, had become out dated or was insufficiently geared to their job, are categorized as differently-educated.

Similarly, respondents are defined as over-skilled if they stated that their knowledge and skills were above the level required by their work. They are defined as under-skilled if they indicated that their knowledge and skills were below the level required by their work. Respondents are skill-matched if they stated that their knowledge and skills were approximately at the level required by their work. Furthermore, the respondents that answered that their knowledge and skills were of a different kind, had become out dated or was insufficiently geared to their job, are categorized as differently-skilled. These skill and educational mismatch variables are made operational by converting them into binary dummy variables. These dummies take a value of 1 when a worker demonstrates the particular type of mismatch in skills or education.

The age of the respondent is quantified in years, representing a continuous measurement. The variable "hours" reflects the average number of hours worked per week by the respondent. The wage of the respondent is measured as the gross monthly wage, encompassing bonuses and other forms of compensation. The variable "female" and "male" are mutually exclusive, since respondents are provided with a choice to identify their gender.

The educational levels of the respondents encompass various categories. These include "barely any education," indicating individuals who have completed only primary school or have no formal education at all; "Pre-highschool," "Highschool," "Further education," which refers to education beyond high school but below university level; "Bachelor," "Masters," and "PhD," denoting different levels of academic attainment.

Furthermore, the sector of work for respondents is categorized into several distinct sectors. The "Primary sector" encompasses occupations related to agriculture, fishery, and mining. The "Secondary sector" represents the manufacturing process that transforms raw materials, often derived from the primary sector, into finished goods. The "Tertiary 1 sector" includes retail trade, catering, and transportation services. The "Commercial sector" pertains to financial and business services. Lastly, the "Tertiary 3 sector" encompasses public and welfare services, such as education, healthcare, government services, and environmental services. A detailed explanation of all the variables used in the mathematical and econometric analyses conducted in this thesis are mentioned in Table 5 in Appendix B.

3.3. *Panel data Methodologies*

Since panel data combines both cross-sectional and time-series data, it offers more variation and more data points, which can improve the precision of estimates. However, due to its structure, several assumptions of standard regression models may be violated, making it inappropriate to use those kind of methods.

For instance, as mentioned previously, standard OLS regression assumes independent observations. In panel data, observations are often correlated across time. A crucial factor that must be considered when dealing with panel data is the presence of unobserved heterogeneity. This refers to the unobserved respondent-specific characteristics that are time-invariant (i.e. constant over time), that can be correlated to other explanatory variables. To overcome these issues, econometricians developed specialized methods for estimating panel data models, such as Pooled OLS, Fixed Effects (FE), and Random Effects (RE) models. These models can help account for the unique characteristics of panel data and produce more accurate and reliable estimates as mentioned previously in the start of the methodology section. These 3 models are further examined and fitted to our framework.

3.3.1. *Pooled OLS Model*

In Pooled OLS estimation, we ignore that there are potentially meaningful differences between the respondents in our panel. The model is estimated as if it were a simple OLS model without distinguishing between cross-sectional and time series observations. This pooling of the data ignores the potential for any specific individual effects and temporal effects. The key assumptions of this method are: Regression coefficients are the same for all respondents, the errors are uncorrelated with the explanatory variables and so the regressors are non stochastic, the error term is independent and identically distributed, i.i.d., with mean 0 and constant variance. Violations of these assumptions can readily result in biased and inconsistent estimates.

Not unlike the models of McGuinness and Sloane (2011) and Allen and Van der Velden (2001), the Pooled OLS model investigated in this thesis is:

$$Y_{it} = \alpha_0 + \alpha_1 L_{it} + \alpha_2 X_{it} + \alpha_3 ED_{it} + \alpha_4 SK_{it} + \varepsilon_{it} \tag{1}$$

where Y is the dependent variable of interest (Hours, Logwage). L is the other dependent variable of interest but now used as an independent variable. For instance if the dependent variable used is $Y = Hours$, then $L = Logwage$; and subsequently if $Y = Logwage$, then $L = Hours$.

Essentially: $L = \begin{cases} Logwage, & \text{if } Y = Hours, \\ Hours, & \text{if } Y = Logwage. \end{cases}$

X is the vector of the control variables: age, female, educational level dummies and sector of work. ED consists of the educational mismatch dummies; being over-educated, under-educated and differently educated. SK consists of the skill mismatch dummies that consist of being over-skilled, under-skilled and differently-skilled.

The parameters $\alpha = (\alpha_0, \alpha_1, \alpha_2, \alpha_3, \alpha_4)$ are the coefficients to be estimated. Note that here $\alpha_2 = (\alpha_{21}, \alpha_{22}, ..., \alpha_{212})$, $\alpha_3 = (\alpha_{31}, \alpha_{32}, \alpha_{33})$, and $\alpha_4 = (\alpha_{41}, \alpha_{42}, \alpha_{43})$ as there are 12 control variables, 3 educational mismatch dummies and 3 skill-mismatch dummies respectively.

To facilitate interpretation of the skill mismatch and educational mismatch variables, respondents with skill and educational matches have been excluded from this model, serving as a reference group. Additionally, respondents from the tertiary 3 sector have been omitted to establish a reference group for the remaining sector indicators. Likewise, respondents with barely any education have been left out to serve as a reference group for the educational level indicators. This exclusion of reference groups mitigates the issue of the dummy variable trap

in our analysis.

ε is the idiosyncratic error term, i runs for all the respondents, hence runs from the first to the last respondent (based on the respondent ID), and $t = 1, 2, 3, 4$ since only the 4 time period of 2019 to 2022 are used. Please note that unless explicitly stated otherwise, all subsequent models in this thesis employ the same variables and definitions as those mentioned in equation (1).

3.3.2. *Fixed Effect Model*

The Fixed Effect method controls for the omitted variables that vary across sections but not across time, and so the FE regression accounts for the time-invariant heterogeneity characteristics of our panel data. Four key assumptions of the FE regression are: ε_{it} has conditional mean 0; all i explanatory variables and all i idiosyncratic error terms are independent and identically distributed (i.i.d.) draws from their joint distribution; large outliers are unlikely; there is no perfect multicollinearity. The type of fixed effect model used is the within-group fixed effect model. The estimation of this model is carried out as follows:
Initially, we have

$$Y_{it} = \alpha_1 L_{it} + \alpha_2 X_i + \alpha_3 ED_{it} + \alpha_4 SK_{it} + \varepsilon_{it}$$

where X, the vector of control variables, is treated as time-invariant in the fixed effect regression. This is because X includes variables that do not change or change predictably within respondents over the study period. The fixed effects model utilizes within-individual variation to estimate the effects of the variables. As such, variables that are constant or nearly constant within individuals over time, such as highest educational attainment or a perfectly predictable variable like age, are absorbed into the individual fixed effects and thus are treated as time-invariant.
Subtracting this equation with the time-average of each of the variables gives

$$(Y_{it} - \bar{Y}_i) = \alpha_1 (L_{it} - \bar{L}_i) + \alpha_3 (ED_{it} - \bar{ED}_i) + \alpha_4 (SK_{it} - \bar{SK}_i) + (\varepsilon_{it} - \bar{\varepsilon}_i).$$

X_i is subtracted with itself since it is time-invariant (i.e. $\bar{X}_i = X_i$) and so drops out of the model. This shows that inclusion of the time-invariant variable X is redundant when using the fixed effect model. The time-average of Y is defined as $\bar{Y}_i = (1/T) \sum_{t=1}^{T} Y_{it}$ where T is the number of years the ith respondent has taken the survey. $\bar{L}_i$, $\bar{SK}_i$, $\bar{ED}_i$ and $\bar{\varepsilon}_i$ are defined similarly. The above equation can be re-written as

$$\tilde{Y}_{it} = \beta_1 \tilde{L}_{it} + \beta_2 \tilde{ED}_{it} + \beta_3 \tilde{SK}_{it} + \tilde{\varepsilon}_{it} \tag{2}$$

where $\tilde{Y}_{it} = (Y_{it} - \bar{Y}_i)$. $\tilde{L}_{it}$, $\tilde{ED}_{it}$, $\tilde{SK}_{it}$ and $\tilde{\varepsilon}_{it}$ are defined similarly. β is the vector of coefficients to be estimated. This approach of carrying out the Fixed Effect estimation is commonly referred to as the "time-demeaned" OLS algorithm. Equation (2) is the within-group Fixed Effect model that is estimated in this paper.

3.3.3. *Random Effect Model*

The Random Effects (RE) model, unlike the Fixed Effects (FE) model, does not eliminate the time-invariant characteristics of the individuals in the panel but considers them as random variables with a certain distribution. Therefore, the RE model can estimate the effects of both time-varying and time-invariant variables, assuming that the individual-specific effects are uncorrelated with the explanatory variables.

4 key assumptions of the Random Effects regression are: The individual-specific effects are random and are uncorrelated with the explanatory variables; the idiosyncratic errors ε_{it} have a mean of zero and are independent and identically distributed (i.i.d.); large outliers are unlikely; there is no perfect multicollinearity.

The estimation of the Fixed Effect model is carried out as follows:
Initially, we have

$$Y_{it} = \alpha_1 L_{it} + \alpha_2 X_i + \alpha_3 ED_{it} + \alpha_4 SK_{it} + u_i + \varepsilon_{it}$$

where u_i is the individual-specific random effect, and ε_{it} is the idiosyncratic error. The time-invariant variable X_i is included in the RE model because this model considers both the within-individual (time-varying) and between-individual (time-invariant) variations. The individual-specific effect u_i is assumed to be uncorrelated with the explanatory variables.

We then apply a Generalized Least Squares (GLS) transformation to this model because the RE regression uses the within-individual and between-individual variances to re-weight the data. The transformed model is:

$$\theta Y_{it} = \alpha_1 \theta L_{it} + \alpha_2 \theta X_i + \alpha_3 \theta ED_{it} + \alpha_4 \theta SK_{it} + \theta \varepsilon_{it}$$

where $\theta = 1 - [\sigma_\varepsilon^2/(\sigma_\varepsilon^2 + T\sigma_u^2)]^{1/2}$, σ_ε^2 is the variance of the idiosyncratic error ε_{it}, σ_u^2 is the variance of the individual-specific effects u_i, and $T = 1$ - 4 is the number of time periods that respondent i answered the survey.

This GLS transformation yields consistent and efficient estimators under the RE assumptions. The above equation can be re-written as:

$$\dot{Y}_{it} = \gamma_1 \dot{L}_{it} + \gamma_2 \dot{X}_i + \gamma_3 \dot{ED}_{it} + \gamma_4 \dot{SK}_{it} + \dot{\varepsilon}_{it} \tag{3}$$

Here $\dot{Y}_{it} = \theta Y_{it}$, $\gamma_1 \dot{L}_{it} = \alpha_1 \theta L_{it}$, $\gamma_2 \dot{X}_i = \alpha_2 \theta X_i$, $\gamma_3 \dot{ED}_{it} = \alpha_3 \theta ED_{it}$, $\gamma_4 \dot{SK}_{it} = \alpha_4 \theta SK_{it}$, and $\dot{\varepsilon}_{it} = \theta \varepsilon_{it}$. Equation (3) is the Random Effect model that is estimated in this paper.

4. DATA

This thesis uses the data from the Longitudinal Internet Studies for the Social Sciences (LISS) panel for the empirical analysis of the impact on skill mismatch on labour force participation. The LISS panel is comprised of a representative sample of individuals and households in the Netherlands. Participants in the panel are selected through a stratified sampling approach, ensuring that the sample is diverse and representative of the Dutch population. This thesis will also assume that LISS panel data is representative of the Dutch population.

The LISS panel is managed by CentERdata, a research institute affiliated with Tilburg University. Panel data is available in the form of longitudinal waves for the years 2008 till 2022. Only the longitudinal waves for the years 2019 till 2022 are used in this paper. Data collection in the LISS panel is primarily conducted through online surveys. To mitigate selection biases, individuals who would otherwise be unable to participate are equipped with a computer and internet access. Incentives in the form of financial compensation are offered to encourage their active engagement. Participants are invited to complete surveys at regular intervals, typically ranging from monthly to annually, depending on the specific study. The surveys cover a wide range of topics, including but not limited to labour market outcomes, education, health, family, and socio-economic characteristics.

4.1. *Data Pre-processing and Data Cleaning*

In this study, the analysis is conducted using four datasets from the 'Work and Schooling' core study, covering the years 2019 to 2022. These specific datasets are chosen because they contain the variables of interest required for evaluating our models and trends. Datasets prior to 2019 lack several variables that are crucial to our analysis, as certain questions were either eliminated or modified. Consequently, utilizing datasets from before 2019 could potentially compromise the reliability and validity of our findings.

Furthermore, since the 'Work and Schooling' core study does not include a gender variable, it is necessary to incorporate the gender information from the 'Background Variables' dataset

of April 2022 to ensure a comprehensive analysis.

Combining the datasets from each year yields a total of 22,109 observations. Working students and respondents aged 66 and above are removed from this dataset. Working students' responses can be subject to significant fluctuations over time and do not accurately represent the active labour force in the Dutch economy. Consequently, the dataset is purged of the 494 working students. Additionally, since the retirement age in the Netherlands is 66, respondents above the age of 66 are removed from the dataset as well. This sample selection criteria reduced our dataset from 22,109 observations to 15,477 observations. In order to accurately assess the effects of labour market mismatch, our data sample is limited to respondents who work a single job. Therefore, individuals who have a side job are deliberately omitted from the dataset, further reducing our dataset to 14,588 observations.

Given that the primary focus of this thesis is to examine the impact of skill mismatch on labour force participation, including unemployed respondents in the analysis would be incongruous. Additionally, the presence of unemployed respondents leads to missing values in most of the variables of interest. Hence, the dataset excludes all the remaining 4,138 unemployed workers observed, further decreasing our dataset to 10,450 observations.

4.1.1. *Dealing with Missing values*

After excluding respondents with a side job from our dataset, we are left with individuals who work only one job. Respondents that had missing values as the answer to whether they had a side job and also stated that they indeed had paid work, were imputed as working only one job. This decision followed the logical assumption that a respondent worked only one job if he stated that he had paid work but did not respond (i.e. reported NA) to the question of whether he worked a side job. However, there were still 58 missing values where respondents did not provide an answer regarding their side job. These 58 respondents were subsequently removed from our dataset. This reduced our dataset from 10,450 observations to 10,392 observations.

It was found that the same 137 respondents answered that they didn't know how well their level of skill matched the skills required in their current job. These 137 respondents were therefore categorized as differently skilled. Similarly, for the respondents who stated that they didn't know how well their educational level matched the education level required in their current job were categorized as differently educated.

We encountered significant issues in our dataset pertaining to our two main dependent variables of interest: hours and wages. Regarding the variable "wage", we found 2,324 observations with missing values for their wage, 1,176 observations stated that they didn't know their wage

and 1,599 preferred not to reveal their wage. These 5,099 observations were purged from our dataset since they did not provide any real information of interest. There were 7 missing values reported for the variable hours and these were subsequently omitted from our dataset as well. Deleting these missing values and unreported values from our dataset reduced it from 10,392 to 5,286 observations. Finally, there remained just 9 missing values in our entire dataset, which were removed as well.

By employing these data selection criteria, we reduce our initial dataset from 22,109 to a final sample of 5,277 observations. This final sample can now be readily used for any and all further empirical analysis.

4.2. *Final Data Sample*

The final sample dataset used in this research comprises of 5,277 observations. These observations collectively constitute an unbalanced panel dataset. The term "unbalanced" in this context refers to the fact that not all respondents participated uniformly across the survey period spanning four years from 2019 to 2022.

This characteristic feature of the unbalanced dataset is primarily attributable to the individual respondent's varying level of engagement with the survey process. In our case, the unbalanced panel dataset consists of 2,786 unique respondents, denoted by $i = 2,786$. These unique respondents have contributed data points to our study over a varying number of years, specifically within a range of $t = 1$ to 4 years. To elaborate further, it means that some participants may have responded to the survey only once over the four-year period, while others may have provided responses every year. Thus, the total number of observations in our unbalanced panel dataset adds up to ($N =$) 5,277.

4.3. *Summary and Descriptive Statistics*

Table 1 presents the summary statistics of the utilized dataset. The respondents exhibit a mean gross wage of 5,777.106 euros, with an average of approximately 33 hours and 10 minutes worked per week. These means are much higher than the expected means for the Netherlands. Notably, a significant portion of the mismatch is observed in the over-educated, differently educated, and over-skilled workers. However, the majority of respondents indicate being well matched with their jobs in terms of both education and skill. Specifically, 52.5% of respondents report having a well matched job based on their education level, while 58.1% report having a well matched job based on their skill level.

We apply the natural logarithm transformation to the wages of the respondents to get "Log-wage" . This transformation helps improve the scaling, symmetry and adhere to the statistical

TABLE I

SUMMARY STATISTICS

	Mean	Standard Deviation	Number of Observations
Wage	5,777.106	137,731.200	5,277
Hours	33.163	13.017	5,277
Age	44.966	12.294	5,277
Female	0.457	0.498	5,277
Mismatches:			
Educationally matched	0.525	0.499	5,277
Over-educated	0.187	0.390	5,277
Under-educated	0.092	0.289	5,277
Differently educated	0.197	0.398	5,277
Over-skilled	0.350	0.477	5,277
Under-skilled	0.022	0.147	5,277
Differently skilled	0.113	0.317	5,277
Skill match	0.581	0.493	5,277
Education Level:			
Pre-highschool	0.043	0.204	5,277
Highschool	0.125	0.330	5,277
Further education	0.547	0.498	5,277
Bachelor	0.030	0.169	5,277
Masters	0.172	0.377	5,277
PhD	0.012	0.110	5,277
Sector of work:			
Primary	0.017	0.131	5,277
Secondary	0.149	0.356	5,277
Tertiary 1	0.146	0.353	5,277
Commerce	0.144	0.351	5,277
Tertiary 3	0.432	0.495	5,277

assumptions made in our models. It also helps normalize the distribution, noticed by the reduction in skewness towards 0 and a decrease in kurtosis, bringing it closer to the value of 3 for a normal distribution as shown in the table below.

SKEWNESS AND KURTOSIS OF WAGES AND LOGWAGE

Variable	Skewness	Kurtosis
Wages	72.42579	5255.866
Logwage	-2.92723	39.44

The Phi coefficient measures the correlation between the dummies representing over-skilled and over-educated workers. The coefficient of 0.43 indicates a moderate positive association between being over-skilled and over-educated among respondents. This suggests that individuals who are over-skilled in their current jobs also tend to be over-educated for those jobs. This finding also implies that including educational and skill mismatch dummies does not lead to significant multicollinearity issues, as the Phi coefficient is typically considered problematic when it exceeds 0.7. Similarly, the Phi coefficient of 0.13 indicates a weaker positive relationship between under-educated and under-skilled workers. It is important to note that the Phi coefficient captures the strength and direction of the association between categorical variables but does not provide information about causality or the magnitude of the relationship.

PHI COEFFICIENTS

	Over-educated	Under-educated	Over-skilled	Under-skilled
Over-educated	1.00			
Under-educated	-0.15	1.00		
Over-skilled	0.43	0.00	1.00	
Under-skilled	-0.06	0.13	-0.11	1.00

5. RESULTS AND DISCUSSION

The findings of the Pooled OLS regression, presented in Table 6 in Appendix C, reveal a noteworthy number of significant estimates. The F-Statistics for the Pooled OLS regression are also significant (p-value < 0.01) for both dependent variables, indicating the overall significance of the regressions. However, the coefficient of determination, R-squared (R^2), remains relatively low at 0.154 and 0.234 when using Hours and Logwage as the dependent variables respectively.

Regarding the first regression with Hours worked as the dependent variable, the Pooled OLS results indicate a significant and negative relationship between over-education and hours worked. Specifically, over-educated workers are estimated to work approximately 1 hour and 25 minutes less than their educationally matched counterpart. The results also suggest that over-skilled workers tend to work more hours compared to skill-matched workers, although this relationship is statistically insignificant. Therefore, it is not possible to draw any meaningful conclusions based on the coefficient associated with over-skilled workers. On the other hand, under-educated and under-skilled workers both exhibit statistically significant effects. Under-educated workers are predicted to work approximately 2 hours and 44 minutes more than their educationally matched counterpart, while under-skilled workers are found to work

approximately 2 hours and 39 minutes less than skill-matched workers.

Moreover, it is important to note that the inclusion of the natural logarithm of gross monthly wages, Logwage, as an explanatory variable yields a significant and substantial effect. This strong positive relationship between a respondent's earnings and the number of hours worked is indicated by the coefficient of 4.600. Additionally, an interesting finding is that female workers exhibit considerably fewer working hours compared to their male counterparts, with female workers working approximately 4 hours and 3 minutes less.

In regards to the second Pooled OLS regression that utilizes Logwage as the dependent variable, the findings indicate that both over-educated and over-skilled workers earn less compared to their well-matched counterparts. The coefficients of -0.132 and -0.061 demonstrate that over-educated workers earn 13.2% less than educationally matched workers, while over-skilled workers earn 6.1% less than their well matched counterparts. The estimates for under-educated and under-skilled workers indicate a slight difference in earnings compared to their educationally or skill-matched counterparts. However, these differences are statistically insignificant, rendering them relatively futile for drawing meaningful conclusions. Women were also seen to earn 27.6% less than men, as per the Pooled OLS results.

Table 7 in Appendix C gives the results of the Fixed Effect regression. Unfortunately, all the estimated coefficients of the FE regression are statistically insignificant. This aligns with the overall insignificance of the Fixed Effect regression itself, as indicated by the F-Statistics having p-value greater than 0.4. One potential explanation for these results is the utilization of an unbalanced dataset. In this study, the time period spans only four years, which is relatively short. Additionally, there is limited response data available for each of the four years, leading to a lack of information for multiple time periods of respondents. These characteristics of an unbalanced panel can introduce issues, particularly if the 'missingness' is not random and if respondents with more time periods differ systematically from those with fewer periods. Such biases can impact the accuracy of the Fixed Effect estimates since the Fixed Effect method relies on within-subject variation. The Random Effect model may offer more robustness to these issues.

Another plausible explanation for these findings could be the omission of time-invariant variables from the model, potentially leading to a significant misspecification. This possibility gains support from the observed significant effects of these time-invariant variables, as demonstrated in Table 8, when estimating our RE model.

The Random Effect regression, as presented in Table 8 in Appendix C, yields statistically significant results. Notably, the RE regression with Logwage as the dependent variable exhibits

the highest R^2 among the regressions carried out in this thesis, indicating a relatively strong explanatory power of the model for Logwage. The majority of the estimates in the RE regression align qualitatively with those obtained in the Pooled OLS regression.

Specifically, in the regression with Hours as the dependent variable, the estimate for over-skilled workers suggests a slight increase in hours worked compared to skill-matched workers, but this estimate is not statistically significant. Similarly, the estimate for under-skilled workers suggests that they work approximately 1 hour and 44 minutes less than skill-matched workers, but this estimate is insignificant as well. The educational mismatch variables do produce statistically significant estimates for this regression. The estimates indicate that over-educated workers work approximately 1 hour and 23 minutes less while under-educated workers work approximately 1 hour and 51 minutes more, in comparison to their educationally matched counterpart.

Lastly, in the regression with Logwages as the dependent variables, the results reveal that over-skilled workers earn 5.8% less than skill-matched workers. Under-skilled workers earn 2.5% less than their skill-matched counterparts, although this relationship lacks statistical significance. Alternatively, over-educated workers earn 6.6% less than their educationally matched counterpart. Again the estimate indicating that under-educated workers earn 0.1% more than educationally matched workers is not statistically significant.

The conclusions drawn in this thesis utilize the estimates found using the Random Effect regressions, since they account for unobserved heterogeneity in our sample panel dataset. The obtained results pertaining to the coefficients associated with the educational mismatch variables provide support for the hypothesis put forth by Sattinger (1993), since we too find that over-educated workers earn less than educationally matched workers, and under-educated workers earn more than educationally matched workers. Moreover, our results extend this understanding by demonstrating that even over-skilled workers earn less than their equally skilled counterparts. Therefore, it could be reasonable to employ Sattinger's job assignment model when integrating skill mismatches as an independent variable in our analysis. Furthermore, our estimate for over-educated workers in the regression model with Logwage as the dependent variable is quantitatively very close to that of Sicherman (1991). Our results indicate that over-educated workers experience a 6.6% less wages, while Sicherman observed a reduction of 7% compared to educationally matched workers. The quantitative proximity of these findings highlights the consistency between our study and Sicherman's research.

The inclusion of Logwage and Hours as independent variables in our analysis, as demonstrated in Table 8, reveals noteworthy results. Particularly when examining the relationship with Hours as the dependent variable, we find a significant positive association. Each unit in-

crease in Logwage corresponds to an increase of 3.81 hours worked by a worker, given all other variables hold constant. These findings underscore the potential impact of earnings on an individual's motivation and labour force participation.

6. CONCLUSION

This study delved into the significant but complex dynamics of skill mismatch on labour force participation, using hours worked and earnings as proxies for such. By incorporating skill-mismatch and educational mismatch indicators, along with relevant control variables, the study crafted a comprehensive model that reflects the nuanced relationship between skills, education, and labour market outcomes. The selected methodology addresses issues commonly encountered in panel data analysis, particularly accounting for unobserved heterogeneity within the sample dataset. The analysis applied a multi-pronged econometric approach, employing Pooled Ordinary Least Squares (OLS), Fixed Effects (FE), and Random Effects (RE) models, on our panel data sample spanning four years. Our findings, albeit nuanced, offer considerable insight into the ramifications of educational and skill mismatch on labour market dynamics.

The results from the Pooled OLS regressions and RE regressions presented substantial evidence regarding the effects of skill and educational mismatches. Utilizing the results of the RE regressions, our findings were largely corroborated by previous research. Over-skilled workers, despite their surplus skill-set, did not enjoy significant benefits in terms of work hours or earnings. They were seen to work more and earn less than their skill-matched counterparts, perhaps pointing to an under utilization of their skills or an over saturated labour market for these skills. The effects of over-education aligned, in part, with the hypotheses of Sattinger (1993), where over-educated workers worked fewer hours and earned less compared to their educationally matched counterparts. Moreover, the lower wage experienced by over-educated workers was very close to that observed by Sicherman (1991), thereby reinforcing the validity and reliability of our findings.

Under-educated workers provided a contrasting perspective, demonstrating increased work hours and marginally higher earnings. This highlights the pivotal role of education in optimizing labour market outcomes. However, for under-skilled workers, the effects were reversed, but lacked statistical significance. This rendered the findings less conclusive.

Drawing from the theory Sattinger (1993), the observed patterns can be interpreted as a result of productivity differences among mismatched and well-matched workers. Over-educated workers, constrained by their job roles and lacking incentives for improvement, have a lower productivity ceiling than their educationally-matched counterparts. In contrast, under-educated workers, who have a higher productivity ceiling, are likely to be more active and effective in

the labour market.

Despite the inability to draw absolute conclusions for all mismatch variables due to the lack of statistical significance, a discernible pattern does emerge. A mismatch, be it due to being over-skilled or over-educated, negatively affects a worker's labour force participation, leading to reduced work hours and earnings. Conversely, workers perceived as under-skilled or under-educated appear to be more actively engaged in their professional environment, logging more work hours, and potentially enjoying higher productivity and earnings.

A salient point of discussion from the analysis is the disparity in work hours and earnings across genders. The data clearly highlighted a gender wage gap, with female workers working fewer hours and earning significantly less than their male counterparts. This finding signals a persistent societal issue demanding comprehensive and sustained remedial measures. The link between logwage and work hours also emerged as an influential factor in our analysis. A significant positive relationship was found, suggesting that an increase in earnings potentially motivates increased labour force participation.

Notwithstanding the valuable insights derived from this study, several limitations should be acknowledged. Firstly, the dataset used spanned only four years and was unbalanced in nature, possibly curtailing the depth of our analysis. Also, despite diligent attempts to account for both skill and educational mismatches alongside relevant controls, there may still be omitted factors influencing labor market outcomes. Moreover, we solely rely on self-reported mismatch rather than taking into account realized mismatch as well, as a result of the limitations present in the LISS survey data. The Fixed Effect regression analysis also did not produce significant findings, potentially due to inherent constraints associated with working with unbalanced data or the exclusion of time-invariant variables. Additionally, a number of the results lacked statistical significance, thereby limiting the capacity to draw robust conclusions from all mismatch variables.

Future research might focus on overcoming these limitations by integrating a more balanced and longitudinal dataset. Refinements could be made in the model specification to account for potential omitted variables. A diverse array of econometric models could be explored, including those impervious to the issues that hampered the Fixed Effect regression in our analysis. Lastly, efforts could be made to validate the findings in various contexts to enhance the generalizability of the conclusions. A deeper understanding of skill and educational mismatches would contribute significantly to policy discussions around labor market dynamics, productivity, and wage equity.

To summarize, our findings mark the importance of matching educational and skill levels with job requirements to optimize labour market outcomes. The complexity of these relationships indicates that there is no 'one-size-fits-all' solution, and that more nuanced and individually tailored approaches might be necessary. The findings of this study could offer valuable insights to policymakers, educators, and employers alike, informing strategies to better align education and skills with job market demands, in an effort to promote more equitable labour market outcomes.

In the broader perspective, this thesis illustrates the multifaceted nature of labour market dynamics, characterized by the interplay between education, skills, gender, earnings, and hours worked. Further research could explore these interrelationships in more depth, integrating other factors such as geographical location, socioeconomic status, and job tenure into the analysis. Our understanding of these dynamics is crucial to address the pressing issues in the labour market, ranging from wage disparities to underemployment and job dissatisfaction.

Through a deeper understanding of these issues, we can move towards a labour market that offers equitable opportunities for all workers, irrespective of their educational qualifications, skill sets, or gender. With this goal in mind, this thesis serves as a stepping stone towards a more comprehensive understanding of labour market dynamics.

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APPENDIX A

TABLE IV: Variable Names and Definitions in the LISS data

Variable Name	*Variable Type*	*Definition*
nomem_encr	Integer	Number of the household member encrypted
_m	Integer	Year and month of the response
geslacht	Categories	Respondents gender. 1 = Male, 2 = Female, 3 = Other.
001	Categories	Respondent has paid work. 0 = No, 1 = Yes.
003	Integer	Respondent's age
005	Categories	see below for definition
006	Categories	see below for definition
402	Categories	see below for definition
577	Integer	Respondent's gross wages per month in the job they currently have, including usual bonuses. -9 = I don't know, -8 = I prefer not to say.
127	Integer	Respondent's average hours worked per week. 999 = I don't know.
031	Categories	see below for definition
549	Categories	Respondent's knowledge and skills are approximately at the level required by work. 0 = No, 1 = Yes, -9 = I don't know.
550	Categories	Respondent's knowledge and skills are higher than the level required by work. 0 = No, 1 = Yes, -9 = I don't know.
551	Categories	Respondent's knowledge and skills are lower than the level required by work. 0 = No, 1 = Yes, -9 = I don't know.
552	Categories	Respondent's knowledge and skills are for another kind of work than for current work. 0 = No, 1 = Yes, -9 = I don't know.
553	Categories	Respondent's knowledge and skills have become outdated because the work has changed. 0 = No, 1 = Yes, -9 = I don't know.
554	Categories	Respondent's knowledge and skills have no relation at all to current work. 0 = No, 1 = Yes, -9 = I don't know.
555	Categories	Respondent's knowledge and skills are insufficiently geared to the work practice. 0 = No, 1 = Yes, -9 = I don't know.
033	Categories	see below for definition
143	Categories	Does the respondent have more than one job. 1 = yes, a sidejob, 2 = yes, my own business or partnership, 3 = no
522	Dummy	Respondent is a work student.

TABLE IV: Variable Names and Definitions in the LISS data

Variable Name	*Variable Type*	*Definition*
523	Dummy	Respondent is long time unemployed.

Variable 005: Highest Level of Education Completed

The variable 005 measured the highest level of education completed by the respondents with a diploma or certificate. The respondents were given these 28 categories to choose from:

1. Did not complete any education
2. Did not complete primary school
3. Primary school
4. Lower and continued special education
5. VGLO (continued lower education)
6. LBO (lower professional education)
7. Lower technical school, household school
8. MULO, ULO, MAVO (lower/intermediate secondary education; US: junior high school)
9. VMBO vocational training program (preparatory intermediate vocational school)
10. VMBO theoretical or combined program (preparatory intermediate vocational school)
11. MMS (intermediate girls' school)
12. HBS (former pre-university education, US: senior high school)
13. HAVO (higher general secondary education; US: junior high school)
14. VWO (pre-university education, US: senior high school)
15. Gymnasium, atheneum, lyceum (types of pre-university education programs)
16. KMBO (short intermediate professional education), VHBO (preparatory higher professional education)
17. MBO professional training program (intermediate professional education) (BOL)
18. MBO professional training program (intermediate professional education) (BBL)
19. MBO-Plus to access HBO, short HBO education (less than two years) (higher professional education)
20. HBO (higher professional education), institutes of higher education, new style
21. Teacher training school
22. Conservatory and art academy
23. Academic education (including technical and economic colleges, former style) bachelor's degree (kandidaats)
24. Academic education (including technical and economic colleges, former style) master's degree (doctoraal)
25. Academic education, bachelor

26. Academic education, master
27. Doctor's degree (Ph.D, including doctoral research program to obtain Ph.D)
28. Other

Variable 006: Highest Level of Education Completed

If the respondents chose category 28 (Other) when answering question 005 (i.e., $if(005 = 28)$), they were asked about their highest level of education using the following categories:

1. None
2. Elementary (comparable to primary education)
3. Middle school (comparable to VMBO, VBO, MAVO; intermediate secondary education)
4. Post-secondary, non-tertiary (comparable to post-MBO; continued intermediate professional education)
5. Tertiary (comparable to HBO, WO; higher professional education, university education)
6. Post-tertiary (comparable to post-academic, including doctorate)

Variable 031: My education...

The variable 031 measures the degree of skill mismatch of the respondent. The respondent is made to choose which of the statements below best describes his situation.
My education is ...

1. is approximately at the level required by my work
2. is higher than the level required by my work
3. is lower than the level required by my work
4. is for another kind of work than for my current work
5. has become outdated because the work has changed
6. has no relation at all to my current work
7. is insufficiently geared to the work practice

The value 99 = I don't know in this case.

Variable 033: If the respondent answered yes for more than 1 dummy in 549 to 555 (i.e. if (sum (549 to 555) > 1)), then they were asked to choose one of the following statements: My knowledge and skills...

1. are approximately at the level required by my work
2. are higher than the level required by my work
3. are lower than the level required by my work
4. are for another kind of work than for my current work
5. have become outdated because the work has changed
6. have no relation at all to my current work
7. are insufficiently geared to the work practice

Again, the value 99 = I don't know here.

Variable 402: Sector of work This variable measure the sector that the respondent worked in. The respondents were given the choice of the 15 categories listed below:

1. agriculture, forestry, fishery, hunting
2. mining
3. industrial production
4. utilities production, distribution and/or trade (electricity, natural gas, steam, water)
5. construction
6. retail trade (including repairs of consumer goods)
7. catering
8. transport, storage and communication
9. financial
10. business services (including real estate, rental)
11. government services, public administration and mandatory social insurances
12. education
13. healthcare and welfare
14. environmental services, culture, recreation and other services
15. other

APPENDIX B

TABLE V: Variable Names and Definitions used in this paper

Variable Name	*Variable Type*	*Definition*
Respondent ID	Integer	Number of the household member encrypted
Date	Integer	Year and month of the response
Female	Dummy	Respondent is Female.
Age	Integer	Respondent's age.
Wage	Integer	Respondent's gross wage per month (based on 577 of LISS)
Logwage	Continuous	The natural logarithm of the the Respondent's gross wage per month.
Hours	Integer	Respondent's average hours worked per week (based on 127 of LISS).
Education Match	Dummy	Respondent's educational level is equally matched with the educational requirement of his job.
Over-educated	Dummy	Respondent's educational level is above the educational requirement of his job.
Under-educated	Dummy	Respondent's educational level is below the educational requirement of his job.
Different Education	Dummy	Respondent's educational level unrelated and different from the educational requirement of his job.
Skillmatch	Dummy	Respondent's skill level matches the skill level required in his job.
Over-skilled	Dummy	Respondent's skill level exceeds the skill level required in his job.
Under-skilled	Dummy	Respondent's skill level is below the skill level required in his job.
Differently skilled	Dummy	Respondent has skill that is unrelated or different to the skills required from his job.
Sector of work	Dummies	see below for definitions of the sectoral dummies.
Education level	Dummies	see below for definitions of the educational dummies.

Sector dummies

The sector dummy variables comprise of the sector that the respondent worked in. These dummies and their definitions are:

Primary sector: The respondent works in the primary sector of the economy. The dummy equals 1 if the respondent answered 1 or 2 for variable 402 of the LISS panel.
Secondary sector: The respondent works in the secondary sector of the economy. The dummy equals 1 if the respondent answered 3, 4 or 5 for variable 402 of the LISS panel.
Tertiary 1: The respondent works in the part of the tertiary sector of the economy that deals in avenues like retail trade, catering and transport. The dummy equals 1 if the respondent answered 6, 7 or 8 for variable 402 of the LISS panel.
Tertiary commerce: The respondent works in the part of the tertiary sector of the economy that deals in financial and commercial services. The dummy equals 1 if the respondent answered 9 or 10 for variable 402 of the LISS panel.
Tertiary 3: The respondent works in the part of the tertiary sector of the economy that deals in public sector service or welfare services. The dummy equals 1 if the respondent answered 11, 12, 13 or 14 for variable 402 of the LISS panel.

Educational level dummies

The educational level dummies comprise of the dummies that measure what the highest level of education the respondent achieved was. These dummies and their definitions are:
Barely any education: The dummy equals 1 if the respondent answered 1 or 2 for variable 005, or 1 for variable 006.
Pre-highschool education: The dummy equals 1 if the respondent answered 3, 4, 5, 6 or 7 for variable 005, or 2 or 3 for variable 006.
Highschool education: The dummy equals 1 if the respondent answered 8, 9, 10, 11, 12, 13 or 14 for variable 005, or 4 for variable 006.
Further education: The dummy equals 1 if the respondent answered 15, 16, 17, 18, 19, 20, 21 or 22 for variable 005, or 5 for variable 006.
Bachelor degree: The dummy equals 1 if the respondent answered 23 or 25 for variable 005, or 6 for variable 006.
Masters degree: The dummy equals 1 if the respondent answered 24 or 26 for variable 005, or 7 for variable 006.
PhD: The dummy equals 1 if the respondent answered 27 for variable 005.

APPENDIX C

TABLE VI
POOLED OLS REGRESSION RESULTS

	Dependent variable:	
	Hours	Logwage
	(1)	(2)
Logwage	4.600***	
	(0.242)	
Hours		0.014***
		(0.001)
Age	−0.084***	0.008***
	(0.014)	(0.001)
Female	−4.043***	−0.276***
	(0.362)	(0.020)
Education Level:		
Pre-highschool	0.713	−0.185***
	(1.032)	(0.057)
Highschool	0.078	−0.061
	(0.787)	(0.043)
Further education	−0.085	0.178***
	(0.660)	(0.036)
Bachelor	−0.648	0.236***
	(1.145)	(0.063)
Masters	3.001***	0.461***
	(0.752)	(0.041)
PhD	2.763*	0.464***
	(1.621)	(0.089)
Sector of Work:		
Primary	4.791***	−0.027
	(1.277)	(0.071)

	Dependent variable:	
	Hours	Logwage
	(1)	(2)
Secondary	1.413***	0.134***
	(0.510)	(0.028)
Tertiary 1	1.108**	−0.075***
	(0.505)	(0.028)
Commerce	0.847*	0.151***
	(0.502)	(0.028)
Mismatches:		
Over-educated	−1.415***	−0.132***
	(0.509)	(0.028)
Under-educated	2.730***	0.046
	(0.610)	(0.034)
Differently educated	1.140**	−0.099***
	(0.489)	(0.027)
Over-skilled	0.343	−0.061***
	(0.396)	(0.022)
Under-skilled	−2.651**	−0.037
	(1.149)	(0.063)
Differently skilled	−0.755	−0.197***
	(0.585)	(0.032)
Constant	0.650	7.183***
	(2.082)	(0.058)
Observations	5,277	5,277
R^2	0.154	0.234
Adjusted R^2	0.151	0.231
F Statistic (df = 19; 5257)	50.350***	84.649***
Note:	*p<0.1; **p<0.05; ***p<0.01	

TABLE VII

FIXED EFFECT REGRESSION RESULTS

	Dependent variable:	
	Hours	Logwage
	(1)	(2)
Logwage	0.631	
	(0.511)	
Hours		0.001
		(0.001)
Over-educated	−0.620	0.009
	(0.804)	(0.032)
Under-educated	−0.756	−0.036
	(0.929)	(0.036)
Differently educated	0.052	−0.002
	(0.729)	(0.029)
Over-skilled	0.629	−0.037
	(0.593)	(0.023)
Under-skilled	−0.711	−0.034
	(1.435)	(0.056)
Differently skilled	−1.320	−0.025
	(0.828)	(0.032)
Observations	5,277	5,277
R^2	0.003	0.002
Adjusted R^2	-1.118	-1.119
F Statistic (df = 7; 2484)	1.036	0.849
Note:	*p<0.1; **p<0.05; ***p<0.01	

TABLE VIII

RANDOM EFFECT REGRESSION RESULTS

	Dependent variable:	
	Hours	Logwage
	(1)	(2)
Logwage	3.810***	
	(0.249)	
Hours		0.008***
		(0.001)
Age	−0.089***	0.008***
	(0.016)	(0.001)
Female	−4.291***	−0.309***
	(0.422)	(0.027)
Pre-highschool	0.430	−0.207***
	(1.085)	(0.061)
Highschool	−0.199	−0.108***
	(0.780)	(0.041)
Further education	−0.179	0.108***
	(0.614)	(0.030)
Bachelor	−0.886	0.152**
	(1.213)	(0.068)
Masters	2.929***	0.356***
	(0.738)	(0.038)
PhD	3.355*	0.422***
	(1.725)	(0.098)
Primary	5.080***	0.124
	(1.495)	(0.092)
Secondary	2.050***	0.126***
	(0.593)	(0.037)
Tertiary 1	1.357**	−0.085**
	(0.584)	(0.036)
Commerce	0.873	0.206***
	(0.584)	(0.036)

	Dependent variable:	
	Hours	Logwage
	(1)	(2)
Mismatches:		
Over-educated	-1.374^{***}	-0.066^{**}
	(0.517)	(0.026)
Under-educated	1.854^{***}	0.001
	(0.621)	(0.032)
Differently educated	0.833^{*}	-0.055^{**}
	(0.487)	(0.025)
Over-skilled	0.234	-0.058^{***}
	(0.396)	(0.020)
Under-skilled	-1.740	-0.025
	(1.085)	(0.052)
Differently skilled	-0.941^{*}	-0.123^{***}
	(0.568)	(0.028)
Constant	7.484^{***}	7.427^{***}
	(2.162)	(0.064)
Observations	5,277	5,277
R^2	0.167	0.722
Adjusted R^2	0.164	0.721
F Statistic	660.795^{***}	721.935^{***}
Note:	$^{*}p<0.1$; $^{**}p<0.05$; $^{***}p<0.01$	